

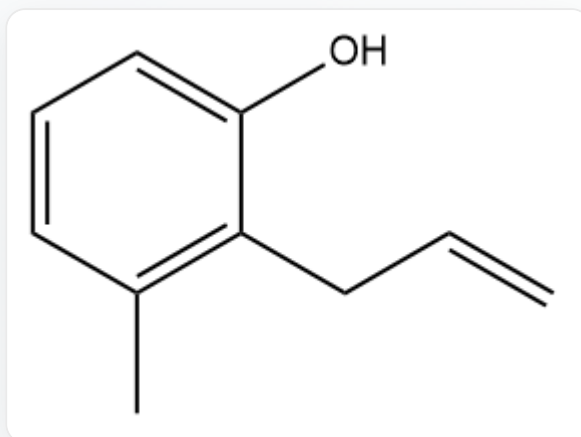
Question

A is an aromatic compound with the molecular formula $C_{10}H_{12}O$. Heating **A** to $200^{\circ}C$ causes isomerization to obtain compound **B**; **B** is treated with dimethyl sulfate to obtain compound **C** ($C_{11}H_{14}O$). **A** undergoes ozonolysis-reduction to yield **D** and formaldehyde; **B** undergoes ozonolysis-reduction to yield **E** and acetaldehyde; while **C** is treated with acidic potassium permanganate solution to obtain *o*-methoxybenzoic acid. Select the structure of **A**.

A. All other options are incorrect.

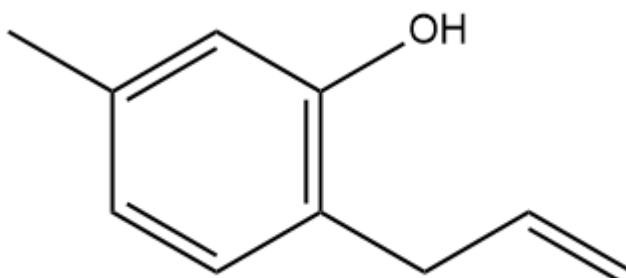
B. There is more than one correct option.

C.



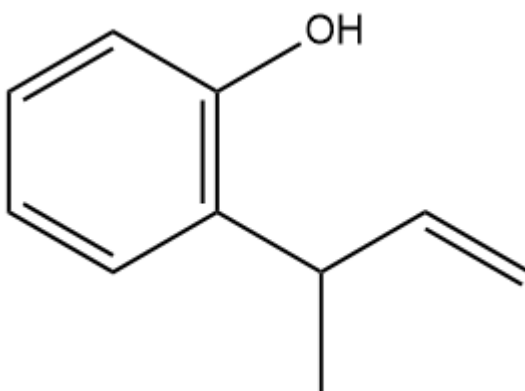
C=CCC1=C(C)C=CC=C1O

D.



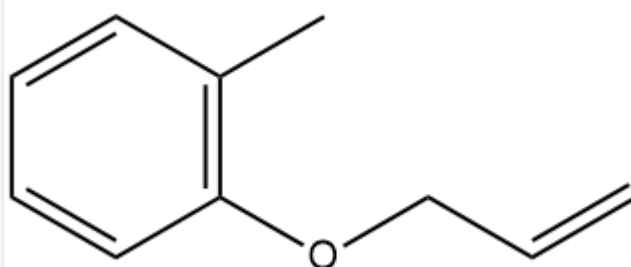
C=CCC1=C(C=C(C)C=C1)O

E.



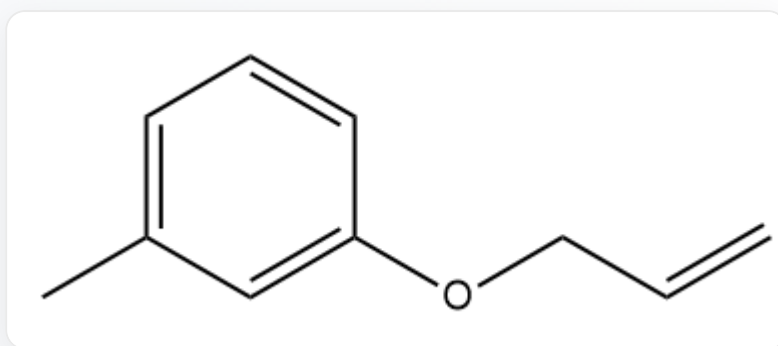
C=CC(C)C1=C(C=CC=C1)O

F.



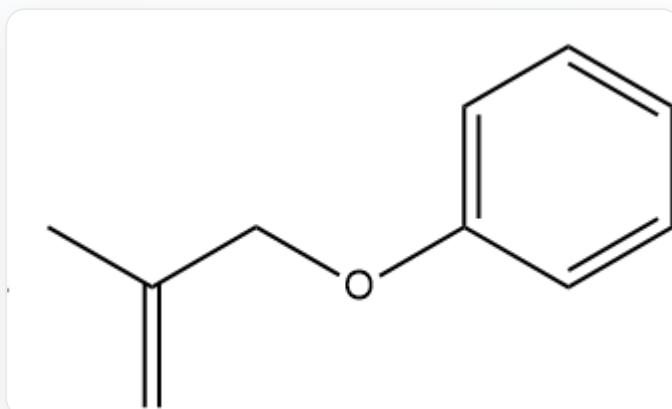
C=CCOC1=C(C)C=CC=C1

G.



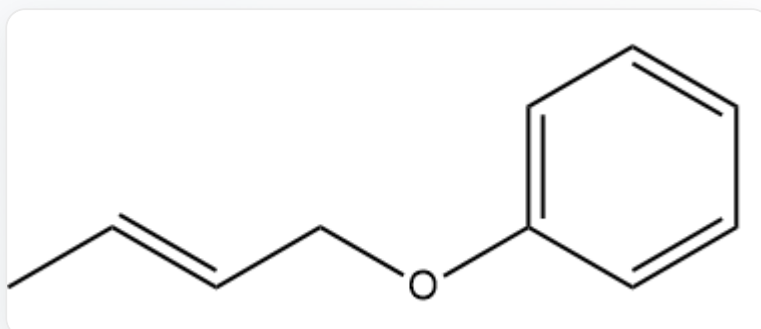
C=CCOC1=CC(=CC=C1)C

H.



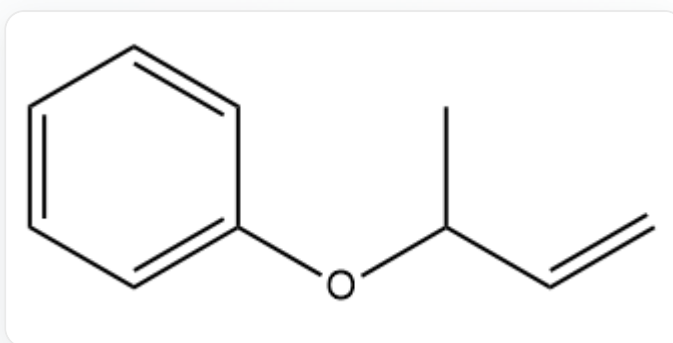
C=C(C)COC1=CC=CC=C1

I.



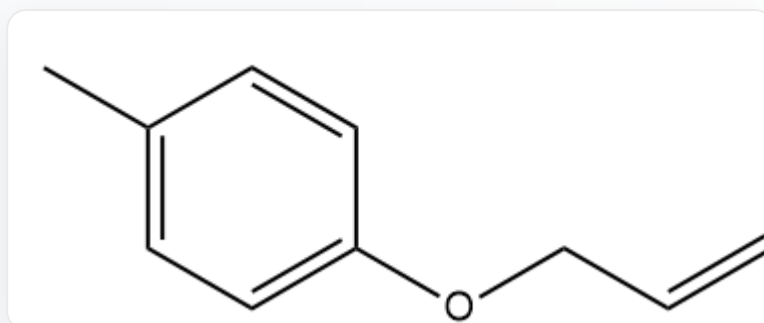
CC=CCOC1=CC=CC=C1

J.



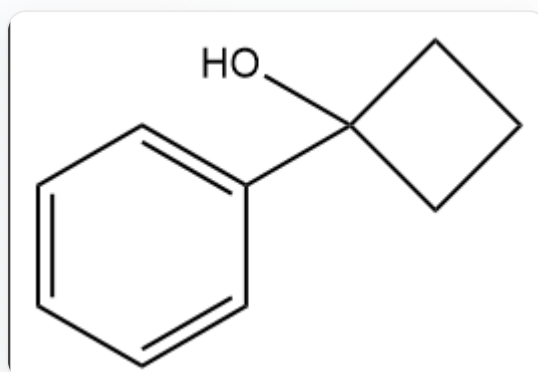
C=CC(C)OC1=CC=CC=C1

K.



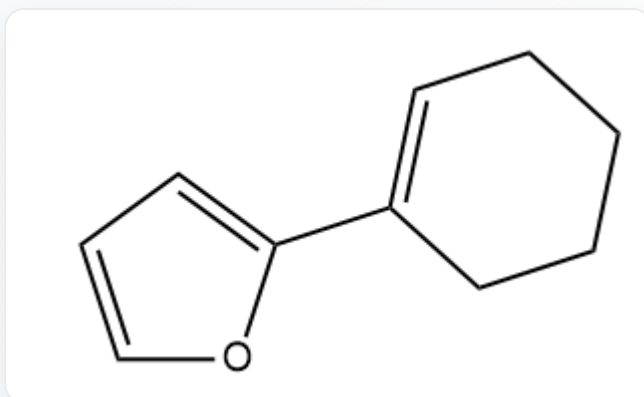
C=CCOC1=CC=C(C)C=C1

L.



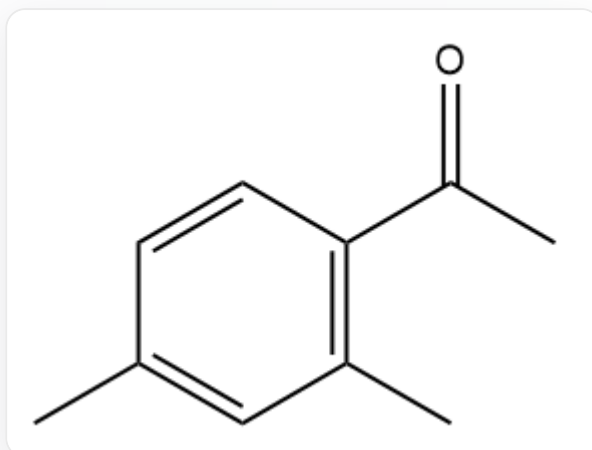
C1=CC=C(C=C1)C2(CCC2)O

M.



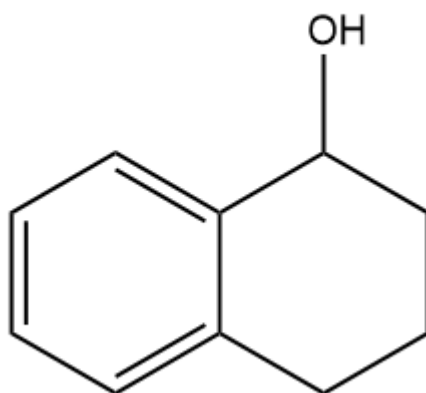
O1C=CC=C1C2=CCCCC2

N.



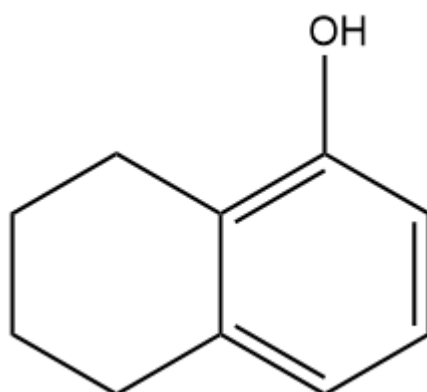
CC1=CC(=C(C=C1)C(=O)C)C

O.



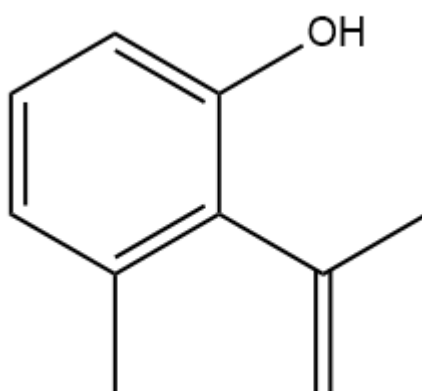
C1=CC=C2C(=C1)CCCC2O

P.



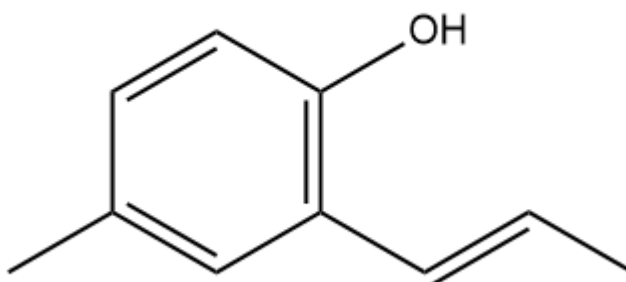
C1CCC2=C(C=CC=C2C1)O

Q.



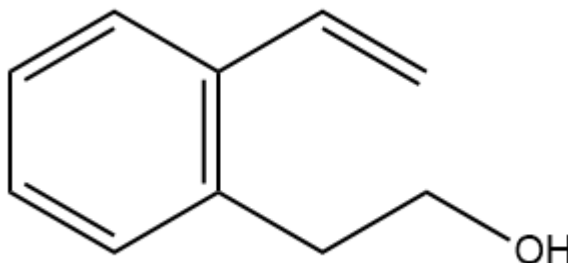
C=C(C)C1=C(C)C=CC=C1O

R.



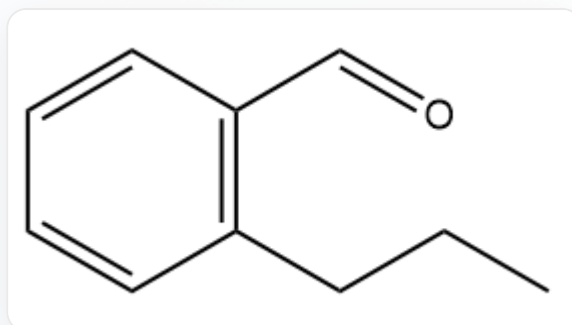
CC=CC1=C(C=CC(=C1)C)O

S.



C=CC1=C(C=CC=C1)CCO

T.



CCCC1=C(C=CC=C1)C=O

Answer

Correct Answer: **J**

Detailed Explanation

A is obtained from an isomerization reaction to yield **B**, then the molecular formula of **B** is also $C_{10}H_{12}O$.

CHECKPOINT

1 PTS

The molecular formula of **B** is $C_{10}H_{12}O$

B undergoes a methylation reaction with dimethyl sulfate to obtain **C**.

CHECKPOINT

1 PTS

B undergoes a methylation reaction to obtain **C**

The molecular formula differs by one CH_2 , and there is only one **O** in **B**, so there is only 1 hydroxyl group in **B**.

CHECKPOINT

2 PTS

There is only 1 hydroxyl group in **B**

C is oxidized by acidic potassium permanganate solution to obtain o-methoxybenzoic acid (COC1=C(C(=O)O)C=CC=C1), indicating that the aromatic ring of **C** is a benzene ring, and it is ortho-disubstituted.

CHECKPOINT

2 PTS

C contains a benzene ring, and it is ortho-disubstituted

The methoxy group will not be oxidized by acidic potassium permanganate solution, while the ortho-hydrocarbon group is oxidized to a carboxyl group, and the chemical formula of the hydrocarbon group is C_4H_7 .

CHECKPOINT

1 PTS

The two substituents on the benzene ring in **C** are methoxy and $-C_4H_7$

Both **A** and **B** can undergo ozonolysis reduction reactions, which contain carbon-carbon double bonds, so the degree of unsaturation of the $-C_4H_7$ group comes from a carbon-carbon double bond.

CHECKPOINT

1 PTS

$-C_4H_7$ group contains one carbon-carbon double bond

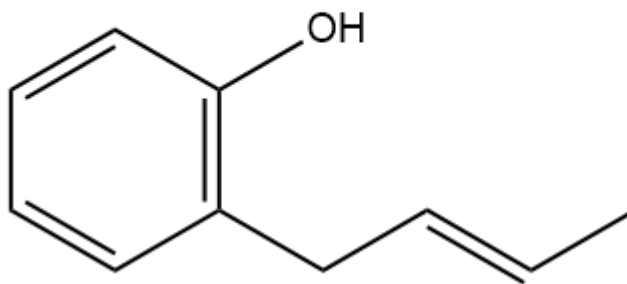
Combining the structure of **B**: o-alkenylphenol, the isomerization reaction of **A** at $200^\circ C$ is likely to be a Claisen rearrangement.

CHECKPOINT

3 PTS

A undergoes Claisen rearrangement to obtain **B**

Therefore, **A**, **B**, **C** all contain methyl-substituted allyl groups. 1-methylallyl and 2-methylallyl are obtained after ozonolysis reduction to obtain formaldehyde; 3-methylallyl (crotyl) is obtained after ozonolysis reduction to obtain acetaldehyde. It can be determined that the structure of **B** is CC=CCC1=C(O)C=CC=C1.



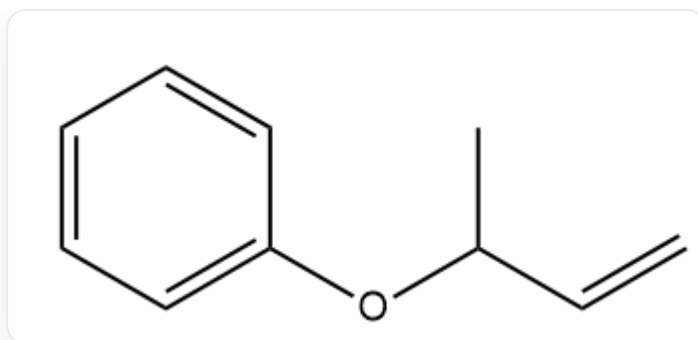
CC=CCC1=C(O)C=CC=C1

CHECKPOINT

2 PTS

The structure of **B** is CC=CCC1=C(O)C=CC=C1

The reverse reaction deduces that the structure of **A** is C=CC(C)OC1=CC=CC=C1.



C=CC(C)OC1=CC=CC=C1

CHECKPOINT

2 PTS

The structure of **A** is C=CC(C)OC1=CC=CC=C1